Simplify Algebraic Expressions

Combine like terms, then find the value of the expression.

Name _____ **Date** _____

Tonight's spark

A team orders 5 shirts and a $7 banner. Later it orders 3 more shirts and a $2 sticker sheet. Each shirt costs $4. What is the total bill?

Guess first: is the total more than $40, or less?

x-terms	Plain numbers
5x	7
3x	2
8x	9
8 × 4 = 32	9

x = 4

Our class model: combine first, then evaluate

Try it

Work it out on paper. Write one answer on each line.

1 **Which term can be combined with 6n?**

A. 2n **B.** 2 **C.** 2m **D.** 6

My answer _____

2 **Simplify 4w + w.**

A. 4w **B.** 5w **C.** 4

My answer _____

3 **Simplify 6c + 3 + 2c + 4.**

A. 15c **B.** 8c + 7 **C.** 8c + 3

My answer _____

4 **What is the value of 5x + 3 when x = 4?**

A. 20 **B.** 35 **C.** 12 **D.** 23

My answer _____

5 **What is the value of 3z + 2 + z + 4 when z = 2?**

A. 12 **B.** 14 **C.** 20

My answer _____

6 **Which expression is already fully simplified?**

A. 4x + 3x **B.** 6 + 1 + 2y **C.** 7x + 3

My answer _____

Talk about it

Which terms are alike? Do the long and the short expression give the same number when x is replaced?

Home mission

Pick x = 4 for the price of one shirt. Ask your student to find 5x + 7 + 3x + 2 two ways: simplify first, or put in 4 first. Do the answers match?

If we get stuck, ask

- Which terms have the same letter part?
- Does this number have a variable attached to it?
- Does the original expression give the same value as the simplified one?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/6-15-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it